Niraiva - OnCoTrack

A Longitudinal Cancer-Signal Intelligence Platform — VC Pitch Document & MVP Product Requirements

Version 2.0 (Consolidated) | August 22, 2026

---

0. The One-Line Pitch

Niraiva turns routine and advanced blood work into a longitudinal cancer-risk radar — closing the loop between an abnormal signal and a confirmed diagnosis, instead of leaving patients with a scary number and no next step.

We are not a lab. We are not a diagnostic test. We are the **intelligence and care-navigation layer** that sits on top of every blood-based cancer signal a person will ever generate — from a $20 CBC to a $950 Galleri-class methylation test — and makes sure nothing abnormal falls through the cracks.

---

1. The Problem

1. **Most cancers are still caught late.** The majority of cancer deaths occur in cancers with no recommended screening test at all, and even where screening exists, adherence is inconsistent and follow-up on abnormal results frequently stalls.
2. **The blood-testing landscape is exploding but fragmented.** A person today might have a PSA from their PCP, a CA-125 from an OB-GYN, a CBC from urgent care, and potentially a Galleri or Guardant test ordered privately — none of these systems talk to each other, and no one is tracking the *trend* across them.
3. **A single lab value is a weak signal; a pattern over time is a strong one.** Clinical literature consistently shows that velocity and multi-marker composites (e.g., PSA velocity, the ROMA index combining HE4 + CA-125) outperform single static values — yet almost no consumer or clinician tool is built to actually track this longitudinally.

4. **New liquid-biopsy technology is arriving faster than the healthcare system's ability to route it.** GRAIL's Galleri MCED test had its FDA Premarket Approval application submitted January 29, 2026, with an FDA advisory committee scheduled to review it September 23, 2026 — this class of test is about to move from a niche LDT to a mainstream, potentially reimbursed product, and there is no category-defining software layer ready to manage the resulting volume of "signal detected, now what?" moments.
5. **Existing tools either overclaim or underdeliver.** Wellness apps that imply they can "detect cancer" from blood work create false reassurance and false alarm risk in equal measure; EHR portals just dump raw lab values on patients with reference ranges and no context.

---

2. Market Opportunity

Segment	**Sizing**	**Source basis**
Liquid biopsy market (broad)	USD 7.05B in 2025, projected to reach USD 22.69B by 2034 (13.91% CAGR)	Precedence Research
Multi-Cancer Early Detection (MCED) testing market	USD 1.92B in 2024, projected to reach USD 7.52B by 2033 (16.5% CAGR)	DataM Intelligence
Alternate MCED forecast	Projected to reach USD 2.86B by 2030 (17% CAGR from 2025)	ResearchAndMarkets

Why now: three converging forces make this the right entry window —

- **Regulatory inflection**: MCED tests are moving from LDT-only to FDA-reviewed status in 2026, which will drive insurance coverage conversations and mainstream ordering volume.

- **Data fragmentation is worsening, not improving**: more test types, more labs, more direct-to-consumer options — the *routing and trend-tracking* problem gets harder every year, which is exactly Niraiva's wedge.
- **No incumbent owns the "software layer."** GRAIL, Guardant, and Exact Sciences are lab/assay companies competing on assay accuracy. None of them are building the neutral, multi-vendor, longitudinal intelligence layer — that's a distinct, defensible business.

---

3. Product Positioning (Critical Framing for Investors and Regulators Alike)

Niraiva does not diagnose cancer. It detects abnormal patterns and missed-screening gaps in blood-based data, and it guarantees that every meaningful signal reaches a clinician and gets closed out — imaging, biopsy, or documented "no further action."

This is not just risk-averse legal framing — it is the actual product wedge. The defensible, monetizable core is:

1. **Longitudinal trend detection** (something no single lab or EHR currently does well), and
2. **Closed-loop follow-up tracking** (which is where patient harm and preventable lawsuits currently concentrate industry-wide).

A "cancer probability score" is scientifically fragile and regulatorily heavy. "We make sure nothing abnormal gets lost" is scientifically honest, technically buildable now, and still extremely valuable.

---

4. Product Architecture: The Biomarker Data Model

To be credible with clinical and regulatory stakeholders, biomarkers are organized into **evidence tiers**, not a flat list. This tiering is itself a core IP asset (a maintained "biomarker knowledge base") and directly drives which biomarkers are allowed to trigger which type of alert.

Tier A — Established Clinical Markers (MVP scope)

Serum protein tumor markers with decades of clinical use, standard CLIA lab availability, and LOINC-coded results:

PSA · CA-125 · CEA · AFP · CA 19-9 · CA 15-3 / CA 27.29 · Beta-hCG · Calcitonin · Thyroglobulin · HE4 · NSE / ProGRP · LDH · Beta-2 microglobulin · Free light chains (kappa/lambda) · SPEP/immunofixation · Chromogranin A · SCC antigen · Cyfra 21-1

Plus **contextual hematology/chemistry**: CBC with differential (and derived ratios — neutrophil-lymphocyte ratio, platelet-lymphocyte ratio), CMP (liver enzymes, renal function, calcium), CRP/ESR, ferritin.

Each marker maps to specific, well-documented cancer associations (e.g., AFP for germ cell tumors and hepatocellular carcinoma; CA-125 for ovarian, endometrial, fallopian tube, lung, breast, and gastrointestinal cancers; CA 19-9 mainly for pancreatic cancer but also colorectal and other GI cancers; CEA for gastrointestinal, cervical, lung, ovarian, breast, and urinary tract cancers), and every marker entry in the knowledge base also documents non-cancer causes of elevation to prevent false-alarm interpretation.

Tier B — Hereditary / Germline & Context-Specific Molecular (Phase 2)

BRCA1/2, Lynch syndrome genes (MLH1/MSH2/MSH6/PMS2/EPCAM), TP53, APC, PALB2, and other panel genes; known-cancer actionable mutations (EGFR, ALK, KRAS, BRAF, HER2, MSI/dMMR status) for patients with an existing diagnosis. This tier answers "what is my inherited risk," which the product must never conflate with "do I currently have cancer."

Tier C — Liquid Biopsy / MCED (Phase 3 integration, not built in-house)

- **ctDNA mutation panels** (Guardant-class)
- **Methylation-based cfDNA / MCED** (Galleri-class): designed to detect cancer-specific methylation patterns shared by many cancer types before symptoms

appear, and when a signal is detected, to predict the cancer signal origin to guide diagnostic evaluation

- **Fragmentomics, copy-number alteration analysis, circulating tumor cells (CTCs)**

Niraiva integrates results from these tests via lab APIs/report parsing — it does **not** attempt to build or validate its own MCED assay. That is a multi-hundred-million-dollar, decade-long undertaking (see GRAIL's trial history) and is explicitly out of scope.

Tier D — Research-Only / Investigational (opt-in research module, not in MVP)

Exosomal RNA/miRNA, autoantibody panels, tumor-educated platelets, metabolomics/lipidomics, immune-cell signatures. Clearly labeled "not for clinical decision-making" if ever surfaced.

Every biomarker record in the knowledge base carries: LOINC code, specimen type, cancer associations, known non-cancer confounders, sensitivity/specificity/PPV/NPV in the relevant population where known, FDA/CLIA/LDT status, and an evidence-tier grade with a review date. This versioned knowledge base is a defensible data asset in itself.

---

5. The MVP — Clearly Scoped

This is the single most important section for a VC conversation: **what ships first, what doesn't, and why.**

5.1 MVP Goal

Prove the core loop — *ingest → trend → flag → route to clinician → track to closure* — using only Tier A biomarkers, which require no new clinical trial, no FDA device submission (if scoped correctly, see §7), and no lab partnership beyond standard interoperability.

5.2 MVP In-Scope Features

Feature	Description
Lab data ingestion	HL7/FHIR import from major lab networks (Quest, LabCorp) + patient portal connections (Apple Health) + PDF/OCR upload with human verification
Longitudinal trend dashboard	Every Tier A biomarker plotted over time against population reference range **and** the patient's own baseline
Velocity/rate-of-change flagging	Detect steep within-range trends (e.g., PSA velocity), not just out-of-range single values
Three-lane result classification	(1) Preventive screening gaps, (2) clinical-concern trends, (3) marker-specific follow-up — never merged into a single "cancer score"
Explainable alerts	Every flag shows exact source values, trend chart, plain-language non-cancer explanations, and a specific recommended next step
Closed-loop referral tracking	Auto-generated referral packet; tracked task from "alert raised" to "clinician disposition documented" — this is the feature that prevents "alert and abandon"
Clinician portal	Multi-patient dashboard, sortable by flag severity, annotation/override tools, EHR export (FHIR)
Screening-gap engine	Cross-reference age/sex/history against USPSTF-style guideline screening schedules (mammography, colorectal, lung CT eligibility, cervical, PSA shared-decision) and flag overdue screening — independent of any biomarker result
Consent center	Granular, revocable consent per data category

5.3 Explicitly Out of MVP Scope (and why)

Excluded	Reason
MCED/ctDNA/methylation test integration	Requires lab partnership agreements and adds Tier C regulatory complexity; validate the core loop on Tier A first, then layer this in Phase 2–3
Autonomous "cancer probability" scoring	Highest regulatory risk (pushes squarely into SaMD/Class II-III territory) and lowest near-term scientific defensibility
Tissue-of-origin prediction	Belongs to the underlying lab test (e.g., Galleri), not to Niraiva — we display it, we don't generate it
Genomic/hereditary risk module	Separate consent, counseling, and GINA-adjacent legal requirements; Phase 2
Research-tier biomarkers (exosomes, metabolomics, etc.)	No near-term clinical utility; premature complexity
Proprietary lab/assay development	Capital- and time-intensive (GRAIL alone has raised and spent well over a billion dollars over a decade to get near FDA review); not the MVP's job

5.4 MVP Success Criteria (what "prove the loop" actually means)

- ≥90% of flagged "clinical concern" or "marker-specific follow-up" alerts reach a documented clinician disposition within a defined SLA (e.g., 14 days) — the core anti-"alert and abandon" metric.
- Measurable increase in completed guideline-based screenings among enrolled users vs. baseline.
- Clinician-reported trust/usefulness score on the explainability UI.
- Zero PHI security incidents; full HIPAA audit pass.

---

6. Clinical & Regulatory Strategy

This is where most "AI detects cancer" startups get into trouble, and it's the section that will get the most scrutiny from serious health-tech VCs.

- **CDS exemption pathway is the target for the MVP.** Under the Cures Act framework, software can potentially avoid full medical-device classification if it lets a physician independently review the basis for the recommendation rather than functioning as a black-box diagnostic output. The MVP's "explainable alert" requirement (§5.2) is designed specifically to keep every feature inside this exemption where possible.
- **IMDRF/FDA risk framework used per-feature, not per-product.** Risk is assessed by (1) how significant the information is to the healthcare decision and (2) how severe the healthcare situation is. A screening-gap reminder is low risk on both axes; a hypothetical future "cancer probability score" would be high risk on both — which is exactly why that feature is deferred, not built into the MVP.
- **Underlying lab tests carry their own regulatory status**, which the product must display transparently rather than implying uniform validation. As of this writing, Galleri's FDA Premarket Approval submission is still under review, built on performance data from the PATHFINDER 2 study and the NHS-Galleri randomized controlled trial — it is not yet an FDA-approved screening test, and the product must say so.
- **HIPAA + genetic-privacy layering.** Standard PHI protections for Tier A data; elevated consent, GINA-awareness, and state genetic-privacy compliance built into the architecture ahead of Phase 2/3, even though not exercised until then.
- **Human-in-the-loop is a non-negotiable architectural constraint**, not a policy statement — no workflow allows a flag to reach a patient without a clinician-review step already logged in the system.
- **Never positioned as a replacement for guideline screening.** This mirrors how Galleri itself is explicitly designed to be used in addition to, and not as a replacement for, guideline-recommended screenings — Niraiva inherits and reinforces that framing at the product level, including for its own screening-gap engine.

---

7. Business Model

Revenue stream	Description	Phase
B2B2C via health systems / employer health plans	License to hospital systems, ACOs, and self-insured employers as a screening-adherence and care-navigation tool — sold on reducing late-stage diagnosis costs and closing quality-measure gaps	MVP
Clinician/practice subscription	Per-provider SaaS fee for the clinician portal (multi-patient dashboard, EHR integration)	MVP
Consumer subscription (freemium)	Free trend-tracking on basic labs; paid tier for premium explainability, family history modeling, and priority care-navigation support	MVP/Phase 2
Lab/test marketplace referral	Facilitation fee when routing a patient to a confirmatory diagnostic test or an MCED test order (compliant, disclosed, non-kickback-structured)	Phase 2–3
De-identified longitudinal data licensing (opt-in, research use)	The multi-year, multi-modal trend dataset becomes increasingly valuable for biomarker validation research over time — a long-term moat, not a Year 1 revenue line	Phase 3+

Why health systems/payers first, not pure consumer: the closed-loop-follow-up problem is a documented cost and liability center for health systems today (missed follow-ups on abnormal results are a leading cause of diagnostic-delay malpractice claims). That gives Niraiva a clear, budget-holding buyer from day one, rather than depending purely on consumer willingness to pay for a wellness app.

---

8. Competitive Landscape

Category	Players	How Niraiva differs
MCED/liquid biopsy assay makers	GRAIL (Galleri), Guardant Health, Exact Sciences, Freenome, DELFI Diagnostics	These are **lab/assay companies** competing on test accuracy and regulatory approval. Niraiva is vendor-neutral software that sits above all of them — it's a complement, not a competitor, and a potential distribution partner for each.
Consumer biomarker/longevity apps	Superpower and similar direct-to-consumer biomarker panels	These focus on broad wellness panels (100+ biomarkers) rather than a clinically governed, closed-loop cancer-signal follow-up system with clinician accountability built in.
EHR-native results portals	Epic MyChart and equivalents	Show raw values with reference ranges; no cross-lab longitudinal trend modeling, no explainability layer, no closed-loop tracking across multiple ordering clinicians.
Traditional oncology navigation platforms	Various care-navigation point solutions	Typically activate *after* a diagnosis; Niraiva's differentiation is pre-diagnosis signal detection plus navigation, closing a gap upstream of where these tools start.

Moat over time: the versioned biomarker evidence knowledge base + the longitudinal, multi-vendor patient trend dataset + closed-loop outcome data (what happened after each flag) compound into a defensible asset that is hard for a single-assay company to replicate, because they only ever see their own test's data.

9. Go-to-Market

1. **Design partner health system(s):** 1–2 pilot health systems or oncology-adjacent primary care networks to validate the closed-loop workflow and generate the first outcome data (Phase 1).
2. **Employer/payer pilots:** self-insured employers and value-based-care ACOs motivated by screening-adherence quality measures (Phase 1–2).
3. **Consumer channel as a funnel, not the initial engine:** direct-to-consumer awareness (content, SEO, partnerships with concierge medicine/longevity clinics) feeding into the clinician-anchored workflow, avoiding a pure-wellness-app growth trap.
4. **MCED/lab partnerships as expansion, not entry:** approach GRAIL, Guardant, and similar labs *after* the core platform has traction — becoming their preferred results-routing and follow-up layer is a natural Phase 2–3 partnership, not a Day 1 dependency.

---

10. Roadmap

Phase	**Scope**	**Primary goal**
Phase 1 (MVP, 0–9 months)	Tier A biomarkers, trend dashboard, explainable alerts, closed-loop referral tracking, clinician portal, screening-gap engine	Prove the core loop with 1–2 design-partner health systems
Phase 2 (9–18 months)	Hereditary/germline risk module, genetic counseling referral pathway, expanded EHR integrations, employer/payer contracts	Expand addressable buyer base, add recurring high-risk-cohort monitoring
Phase 3 (18–30 months)	MCED/ctDNA lab integrations (Galleri-class, Guardant-class) under clinician-ordered, consented workflows only	Become the neutral routing/follow-up layer for the

Phase	Scope	Primary goal
		emerging MCED category as it scales post-FDA-review
Phase 4 (30+ months)	Prospective outcome studies comparing Niraiva-assisted pathways vs. usual care (time-to-diagnosis, stage-shift, avoidable testing)	Build the clinical evidence base for reimbursement conversations and potential future SaMD-tier features

---

11. Key Risks & Mitigations

Risk	Mitigation
Regulatory reclassification as SaMD	MVP feature set deliberately engineered to qualify for CDS exemption; regulatory counsel engaged pre-launch; formal FDA Q-Submission process used before any future risk-scoring feature
False positives causing patient harm/anxiety	Three-lane result architecture, mandatory non-cancer-cause disclosure on every alert, no single-weak-marker alerts, human-in-the-loop clinician review required
Health-system sales cycle length	De-risked by starting with 1–2 design partners and a subscription-plus-pilot structure rather than requiring full enterprise procurement pre-Series A
Dependence on third-party lab data quality	Preserve original lab/method/units/date at ingestion; do not normalize across incompatible assays as if equivalent
Category timing risk (MCED regulatory status still evolving)	MVP does not depend on MCED tests at all — Tier A biomarkers alone justify the core product; MCED integration is upside, not dependency

Risk	Mitigation
Competitive response from assay makers building their own software layer	Vendor-neutral positioning and multi-lab data aggregation is structurally hard for a single-assay company to replicate credibly

---

12. The Ask

(Fill in with actual round specifics — structure below is ready for your numbers.)

- **Raise:** [Seed / Series A amount]
- **Use of funds:** engineering (ingestion pipeline, trend engine, clinician portal), regulatory/clinical advisory board, 1–2 design-partner pilot execution, HIPAA/security compliance buildout
- **Milestones this raise achieves:** MVP live with design partners, ≥90% closed-loop follow-up rate demonstrated, signed LOIs with 1–2 additional health systems or payers, regulatory strategy validated with outside counsel/FDA pre-submission meeting

---

Appendix A — Full Biomarker Reference (Tier A, MVP Scope)

Biomarker	Primary Cancer Association(s)	Common Non-Cancer Elevators
PSA	Prostate	BPH, prostatitis, recent ejaculation, catheterization
CA-125	Ovarian, endometrial, fallopian tube, lung, breast, gastrointestinal	Endometriosis, menstruation, pregnancy, pelvic inflammatory disease

Biomarker	Primary Cancer Association(s)	Common Non-Cancer Elevators
CEA	Gastrointestinal, cervical, lung, ovarian, breast, urinary tract	Smoking, inflammatory bowel disease
AFP	Germ cell tumors, hepatocellular carcinoma	Pregnancy, liver disease/cirrhosis
CA 19-9	Mainly pancreatic, also colorectal and other GI cancers	Biliary obstruction, pancreatitis (absent in Lewis-antigen-negative individuals)
CA 15-3 / CA 27.29	Breast cancer	Benign breast disease, liver disease
Beta-hCG	Gestational trophoblastic disease, germ cell tumors, choriocarcinoma	Pregnancy
Calcitonin	Medullary thyroid carcinoma	C-cell hyperplasia
Thyroglobulin	Thyroid cancer surveillance	Any residual normal thyroid tissue
HE4	Ovarian cancer	Renal impairment
NSE / ProGRP	Small-cell lung cancer, neuroendocrine tumors	Chronic kidney disease
LDH	Lymphoma, germ cell tumors, melanoma (prognostic)	Hemolysis, tissue damage broadly — highly nonspecific
Beta-2 microglobulin	Multiple myeloma, lymphoma, CLL	Renal impairment

Biomarker	Primary Cancer Association(s)	Common Non-Cancer Elevators
CBC + differential	Leukemia/lymphoma pattern flags	Infection, inflammation, medication effects

All entries above are for pattern-recognition and trend-tracking purposes only and require clinician interpretation in clinical context; none is validated as a standalone population-screening test for asymptomatic individuals.

This document consolidates and supersedes prior draft versions. It is a product/business planning draft, not medical, legal, or regulatory advice. Regulatory classification must be confirmed with qualified counsel and, where applicable, directly with the FDA before development of any risk-scoring or diagnostic-adjacent feature.